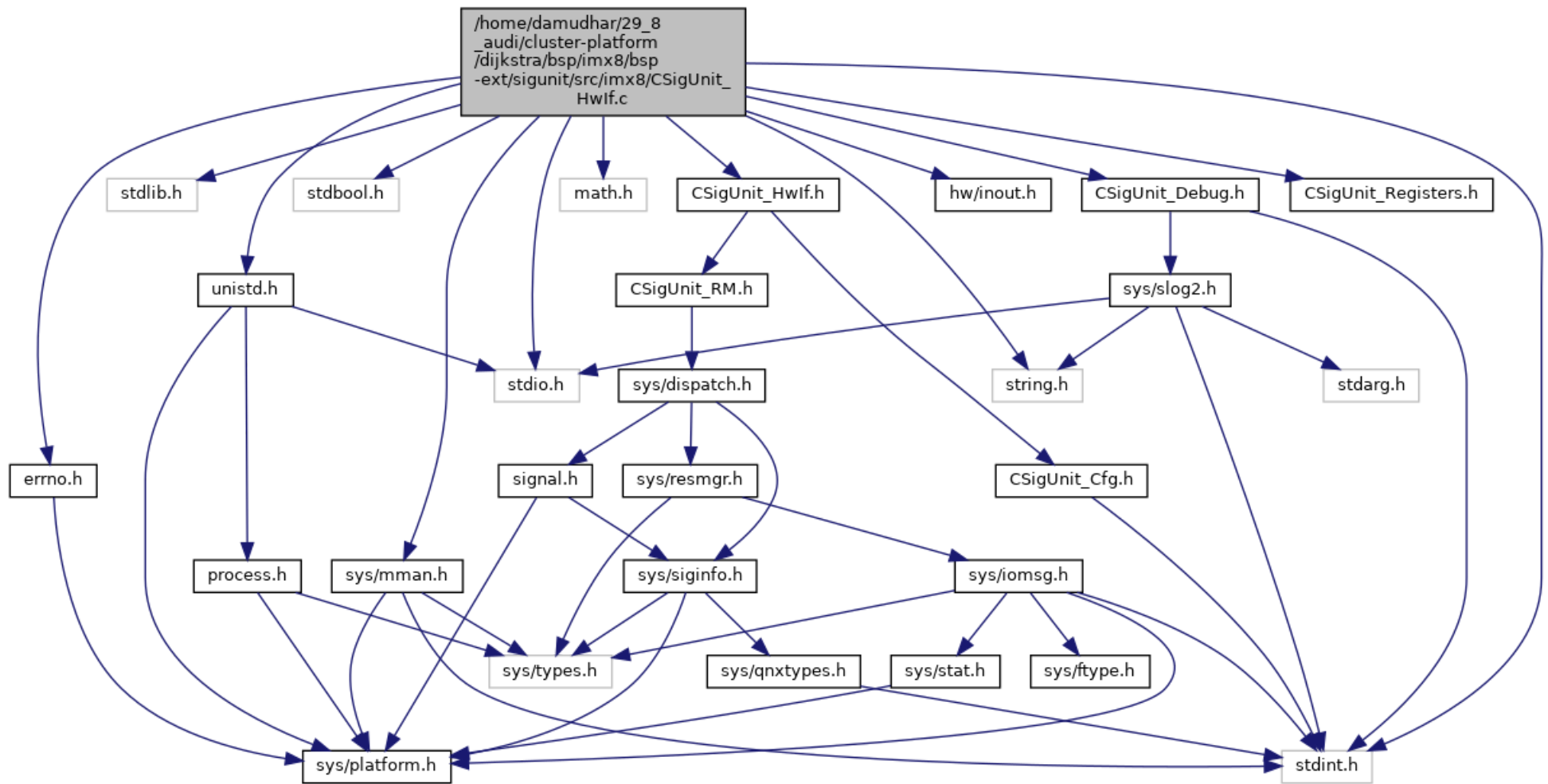


CSigUnit_Hwlf.c File Reference

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <stdbool.h>
#include <unistd.h>
#include <stdint.h>
#include <math.h>
#include <errno.h>
#include <sys/mman.h>
#include <hw/inout.h>
#include "CSigUnit_HwIf.h"
#include "CSigUnit_Registers.h"
#include "CSigUnit_Debug.h"
```

Include dependency graph for CSigUnit_Hwlf.c:



Functions

void **sigunit_set_port32** (uintptr_t reg_addr, uint32_t data)

Function to write a register. [More...](#)

uint32_t **sigunit_get_port32** (uintptr_t reg_addr)

Function to read a register. [More...](#)

static void **sigunit_lock** (const **ts_sigunit_dev_t** *p_sig_lock_dev_t_SP)

Function to lock signature unit. [More...](#)

static uint32_t **sigunit_get_eval_win_status** (const **ts_sigunit_dev_t** *p_siggetevalstatusdev_t_SP, uint8_t eval_win_id_sts_U8)

Function to get evaluation window status. [More...](#)

static uint32_t **sigunit_get_win_crc_r** (const **ts_sigunit_dev_t** *p_sigcrc_r_dev_t_SP, uint8_t eval_win_id_crc_r_U8)

Function to get crc value for red region for user passed eval window. [More...](#)

static uint32_t **sigunit_get_win_crc_g** (const **ts_sigunit_dev_t** *p_sigrcr_g_dev_t_SP, uint8_t eval_win_id_crc_g_U8)

Function to get crc value for green region for user passed eval window. [More...](#)

static uint32_t **sigunit_get_win_crc_b** (const **ts_sigunit_dev_t** *p_sigcrc_b_dev_t_SP, uint8_t eval_win_id_crc_b_U8)

Function to get crc value for blue region for user passed eval window. [More...](#)

static void **sigunit_unlock** (const **ts_sigunit_dev_t** *p_sig_unlock_dev_t_SP)

Function to unlock signature unit. [More...](#)

int32_t **sigunit_initialize** (const **ts_sigunit_dev_t** *pDev, **ts_SSigUnitCfg_t** *p_SigUnitCfg_SP)

Function to initialize signature unit. [More...](#)

int32_t **sigunit_shutdwon** (const **ts_sigunit_dev_t** *pDev)

Function to shutdown signature unit. [More...](#)

int32_t **sigunit_eval_win_setup** (const **ts_sigunit_dev_t** *pDev, **ts_SSigUnitWinCfg_t** *p_SigUnitWinCfg_SP)

Function to setup and evaluation window. [More...](#)

int32_t **sigunit_eval_win_start** (const **ts_sigunit_dev_t** *pDev, uint8_t eval_win_id_U8)

Function to start and evaluation window. [More...](#)

int32_t **sigunit_eval_win_stop** (const **ts_sigunit_dev_t** *pDev, uint8_t eval_win_id_U8)

Function to stop evaluation window by disabling continous mode. [More...](#)

int32_t **sigunit_eval_win_status** (const **ts_sigunit_dev_t** *pDev, **ts_SSigUnitGetWinStatus_t** *p_SigUnitWinStatus_SP)

Function to get status of evaluation window. [More...](#)

Variables

static **ts_SSigUnitEvalWinReg_t** **win_setup_reg_sa** []

static uintptr_t **sig_crc_red_a** []

static uintptr_t **sig_crc_green_a** []

static uintptr_t **sig_crc_blue_a** []

static uint32_t **sig_action_crtl** [3] = { 0x0, 0x10002, 0x20002}

static uint32_t **sig_alpha_crtl** [3] = { 0x0, 0x100, 0x200}

slog2_buffer_set_config_t **slog_config**

slog2_buffer_t **slog_handle**

ts_SSigUnitCheck_t **sigunit_flag** [SIG_UNIT_MAX]

ts_SSigUnitEvalWinCheck_t **evalwin_flag** [SIG_UNIT_EVAL_WIN_MAX]

Function Documentation

◆ sigunit_eval_win_setup()

```
int32_t sigunit_eval_win_setup ( const ts_sigunit_dev_t * p_sigsetupdev_SP,  
                                ts_SSigUnitWinCfg_t * p_SigUnitWinCfg_SP  
                                )
```

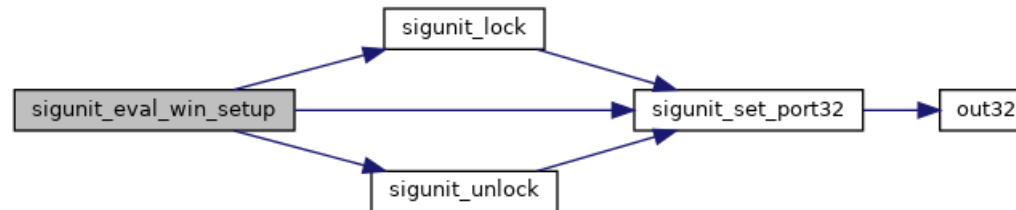
Function to setup and evaluation window.

Returns

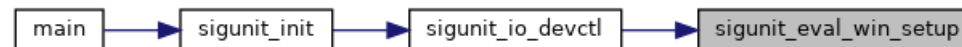
I_winsetup_status_S32

requirement ID 744947,744958

Here is the call graph for this function:



Here is the caller graph for this function:



◆ sigunit_eval_win_start()

```
int32_t sigunit_eval_win_start ( const ts_sigunit_dev_t * p_sigevalstartdev_SP,  
                                uint8_t                 eval_win_id_win_start_U8  
                                )
```

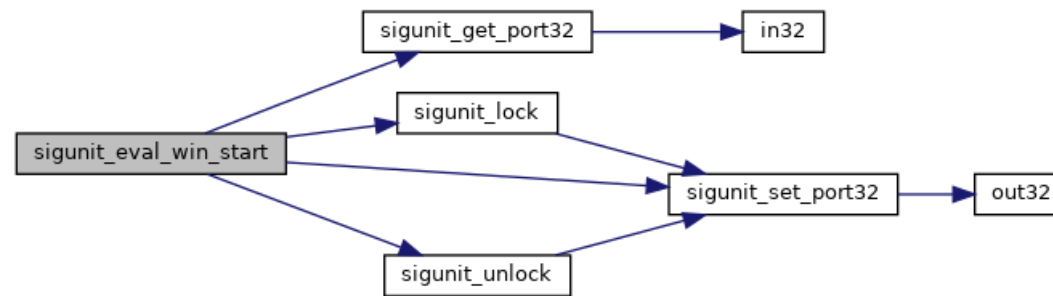
Function to start and evaluation window.

Returns

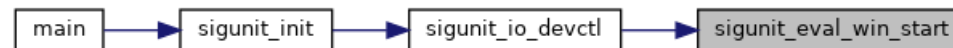
I_winstart_status_S32

requirement ID 744947,744959

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_eval_win_status()`

```
int32_t sigunit_eval_win_status ( const ts_sigunit_dev_t *    p_sigevalstatusdev_SP,
                                ts_SSigUnitGetWinStatus_t * p_SigUnitWinStatus_SP
                                )
```

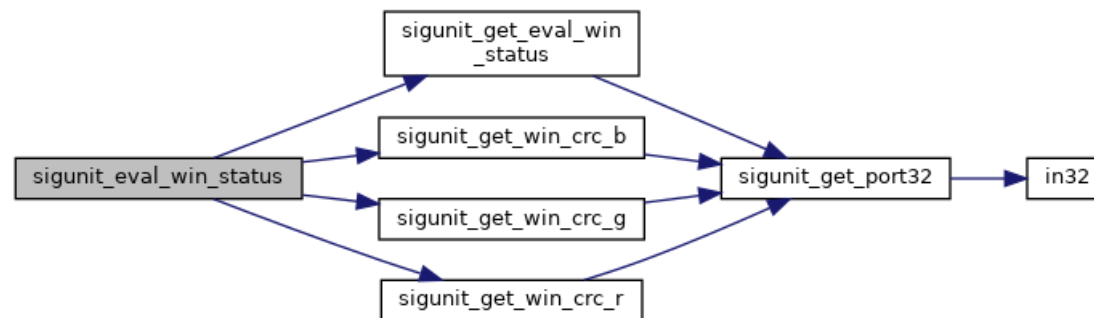
Function to get status of evaluation window.

Returns

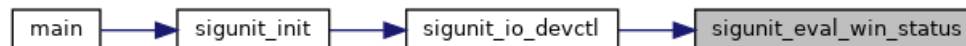
l_winstatus_status_S32

requirement ID 744947,744961

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_eval_win_stop()`

```
int32_t sigunit_eval_win_stop ( const ts_sigunit_dev_t * p_sigevalstopdev_SP,
                               uint8_t                eval_win_id_win_stop_U8
                               )
```

Function to stop evaluation window by disabling continuous mode.

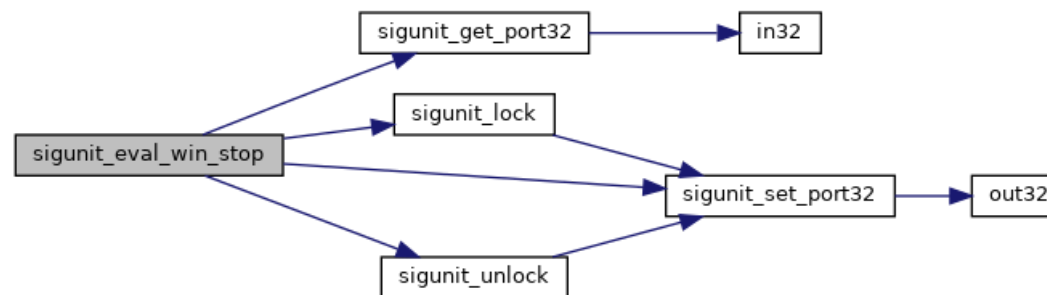
Returns

I_winstop_status_S32

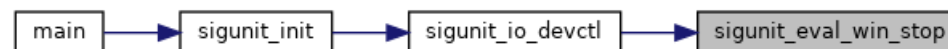
requirement ID 744947,744960

sigunit_set_port32(pDev->regbase + (uintptr_t) DISENG_SIG_CONTINUOUSMODE, (uint32_t) 0u);

Here is the call graph for this function:



Here is the caller graph for this function:



◆ sigunit_get_eval_win_status()

```
static uint32_t sigunit_get_eval_win_status ( const ts_sigunit_dev_t * pDev,
                                             uint8_t                eval_win_id_U8
                                             )
```

static

Function to get evaluation window status.

Returns

I_status_U32

requirement ID 744947,744961

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_get_port32()`

```
uint32_t sigunit_get_port32 ( uintptr_t reg_addr )
```

Function to read a register.

Parameters

offset Register address.

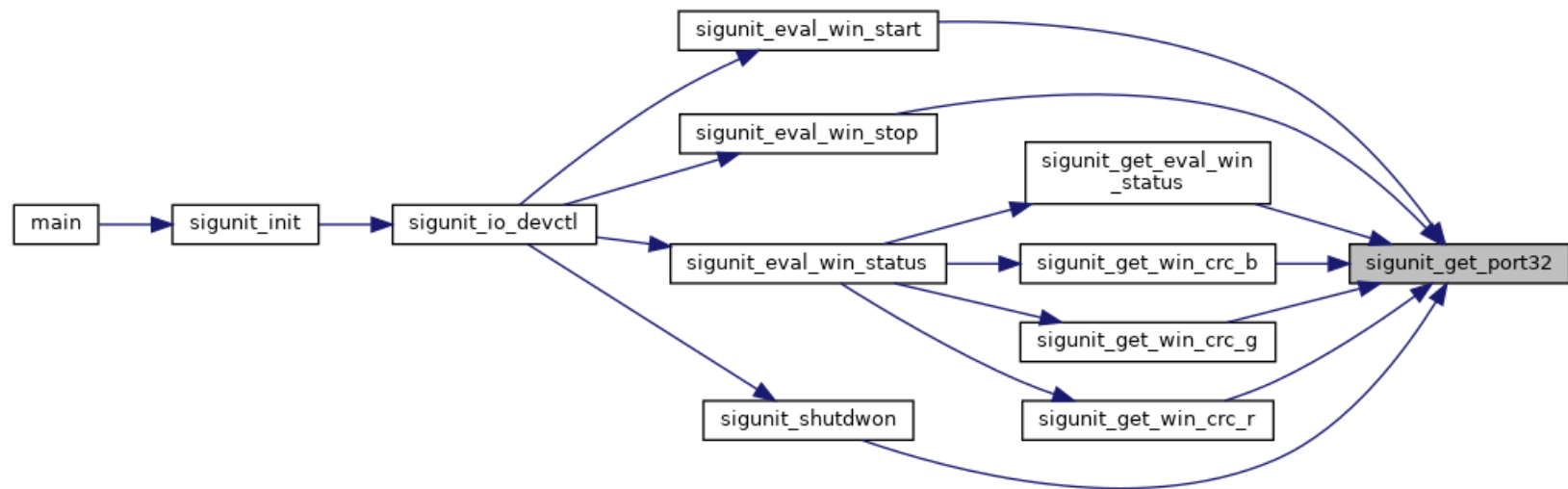
data Data to be read from a register.

requirement ID 744947

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_get_win_crc_b()`

```
static uint32_t sigunit_get_win_crc_b ( const ts_sigunit_dev_t* pDev,  
                                       uint8_t               eval_win_id_U8  
                                       )
```

static

Function to get crc value for blue region for user passed eval window.

Returns

l_ret_value_crc_b_U32

requirement ID 744947,744961

Here is the call graph for this function:



Here is the caller graph for this function:



◆ sigunit_get_win_crc_g()

```
static uint32_t sigunit_get_win_crc_g ( const ts_sigunit_dev_t* pDev,  
                                       uint8_t               eval_win_id_U8  
                                       )
```

static

Function to get crc value for green region for user passed eval window.

Returns

l_ret_value_crc_g_U32

requirement ID 744947,744961

Here is the call graph for this function:



Here is the caller graph for this function:



◆ sigunit_get_win_crc_r()

```
static uint32_t sigunit_get_win_crc_r ( const ts_sigunit_dev_t * pDev,  
                                     uint8_t                eval_win_id_U8  
                                     )
```

static

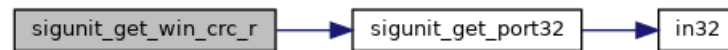
Function to get crc value for red region for user passed eval window.

Returns

l_ret_value_crc_r_U32

requirement ID 744947,744961

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_initialize()`

```
int32_t sigunit_initialize ( const ts_sigunit_dev_t * p_siginit_dev_SP,  
                           ts_SSigUnitCfg_t *      p_SigUnitCfg_SP  
                           )
```

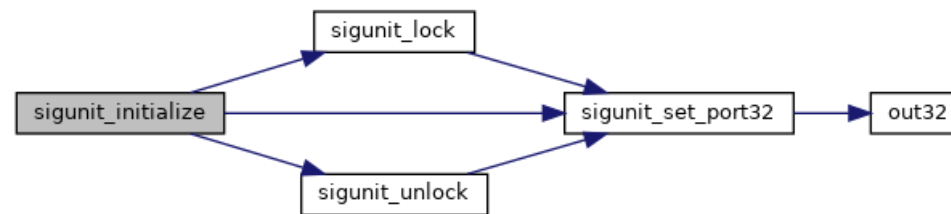
Function to initialize signature unit.

Returns

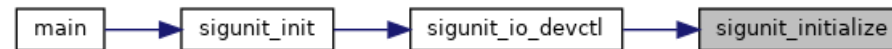
I_sigInit_status_S32

requirement ID 744947,744957

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_lock()`

```
static void sigunit_lock ( const ts_sigunit_dev_t* pDev )
```

static

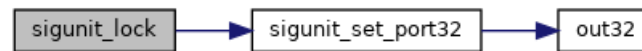
Function to lock signature unit.

Returns

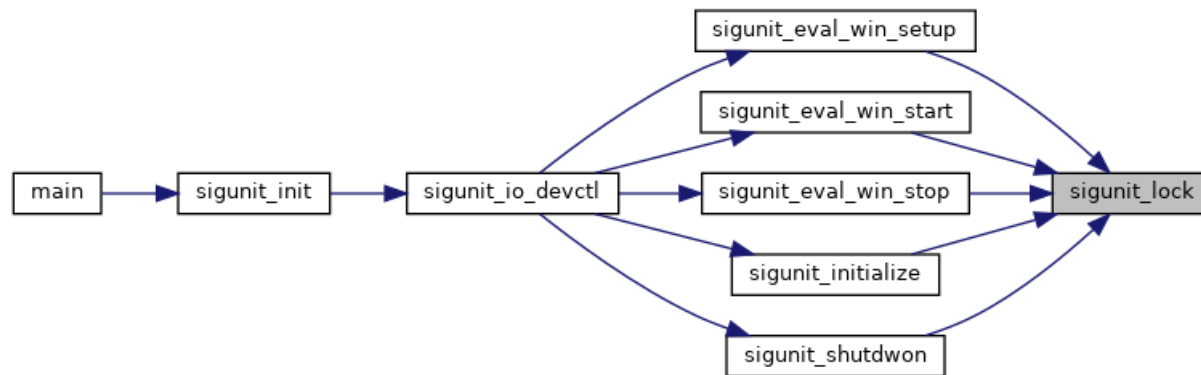
void

requirement ID 744947

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_set_port32()`

```
void sigunit_set_port32 ( uintptr_t reg_addr,
                        uint32_t data
                      )
```

Function to write a register.

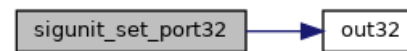
Parameters

offset Register address.

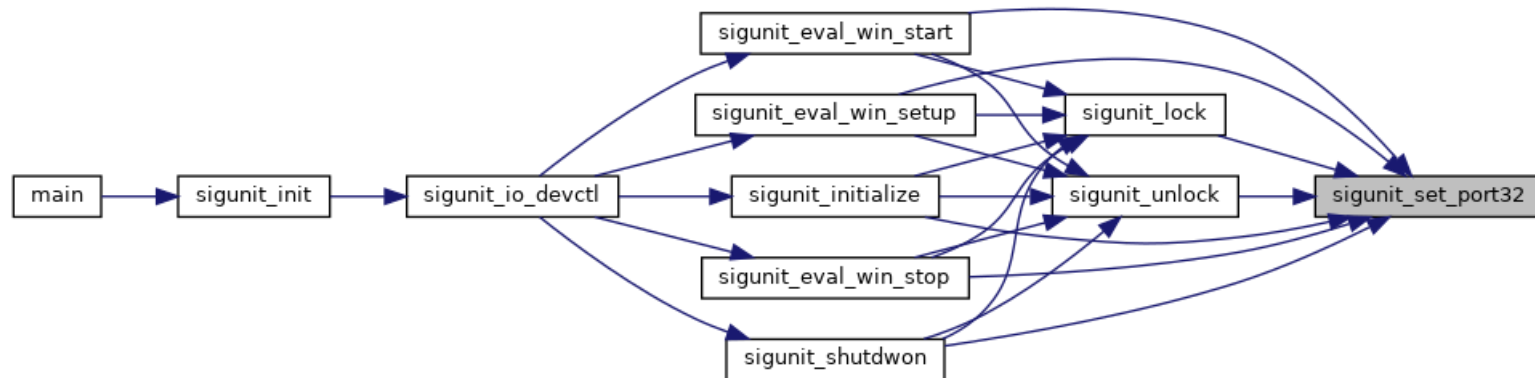
data Data to be written into a register.

requirement ID 744947

Here is the call graph for this function:



Here is the caller graph for this function:



♦ sigunit_shutdwon()

```
int32_t sigunit_shutdwon ( const ts_sigunit_dev_t * pDev )
```

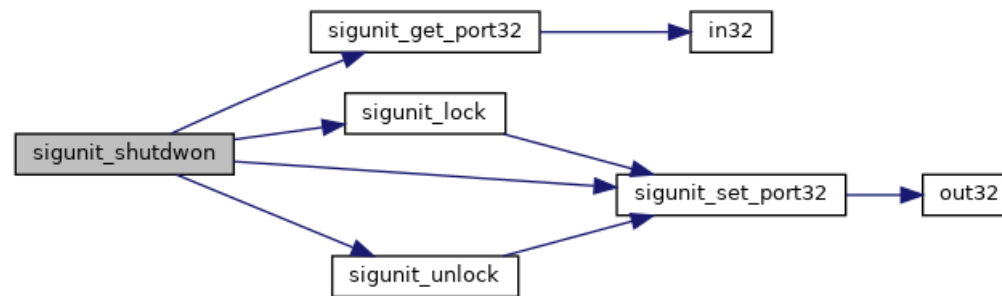
Function to shutdown signature unit.

Returns

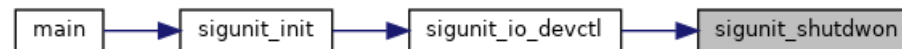
I_sigshutdown_status_S32

requirement ID 744947,744953

Here is the call graph for this function:



Here is the caller graph for this function:



◆ `sigunit_unlock()`

```
static void sigunit_unlock ( const ts_sigunit_dev_t * pDev )
```

static

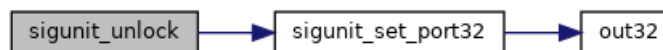
Function to unlock signature unit.

Returns

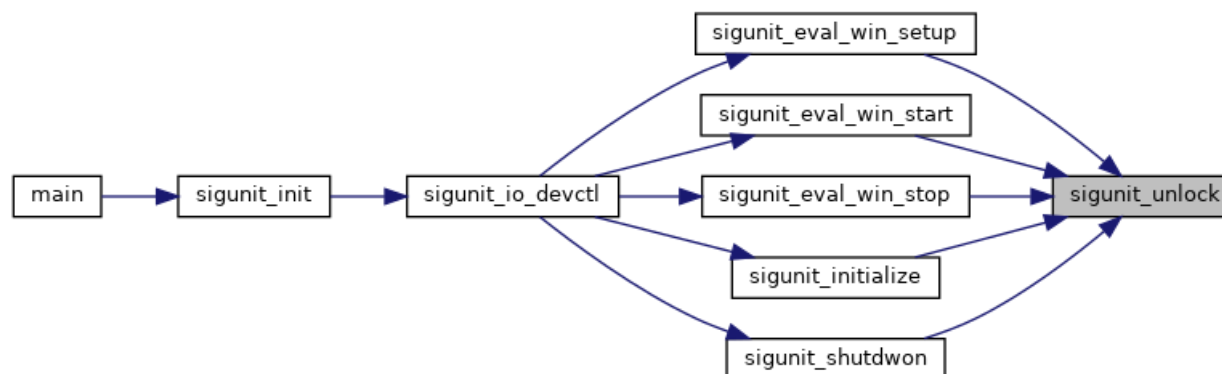
void

requirement ID 744947

Here is the call graph for this function:



Here is the caller graph for this function:



Variable Documentation

◆ evalwin_flag

ts_SSigUnitEvalWinCheck_t evalwin_flag[SIG_UNIT_EVAL_WIN_MAX]

◆ sig_action_crtl

uint32_t sig_action_crtl[3] = { 0x0, 0x10002, 0x20002}

static

uint32_t sig_alpha_crtl[3] = { 0x0, 0x100, 0x200}

static

◆ sig_alpha_crtl

◆ sig_crc_blue_a

uintptr_t sig_crc_blue_a[]

static

Initial value:

```
= { DISENG_SIG_SIGCRCBLUE0, DISENG_SIG_SIGCRCBLUE1, DISENG_SIG_SIGCRCBLUE2,  
    DISENG_SIG_SIGCRCBLUE3, DISENG_SIG_SIGCRCBLUE4, DISENG_SIG_SIGCRCBLUE5,  
    DISENG_SIG_SIGCRCBLUE6, DISENG_SIG_SIGCRCBLUE7  
}
```

◆ sig_crc_green_a

uintptr_t sig_crc_green_a[]

static

Initial value:

```
= { DISENG_SIG_SIGCRCGREEN0, DISENG_SIG_SIGCRCGREEN1, DISENG_SIG_SIGCRCGREEN2,  
    DISENG_SIG_SIGCRCGREEN3, DISENG_SIG_SIGCRCGREEN4, DISENG_SIG_SIGCRCGREEN5,  
    DISENG_SIG_SIGCRCGREEN6, DISENG_SIG_SIGCRCGREEN7  
}
```

◆ sig_crc_red_a

uintptr_t sig_crc_red_a[]

static

Initial value:

```
= {DISENG_SIG_SIGCRCRED0, DISENG_SIG_SIGCRCRED1, DISENG_SIG_SIGCRCRED2,  
    DISENG_SIG_SIGCRCRED3, DISENG_SIG_SIGCRCRED4, DISENG_SIG_SIGCRCRED5,  
    DISENG_SIG_SIGCRCRED6, DISENG_SIG_SIGCRCRED7  
}
```

◆ sigunit_flag

ts_SSigUnitCheck_t sigunit_flag[SIG_UNIT_MAX]

◆ slog_config

slog2_buffer_set_config_t slog_config

◆ slog_handle

slog2_buffer_t slog_handle

◆ win_setup_reg_sa

ts_SigUnitEvalWinReg_t win_setup_reg_sa[]

static

Initial value:

```
= {  
    { DISENG_SIG_EVALCONTROL0, DISENG_SIG_EVALUPPERLEFT0, DISENG_SIG_EVALLOWERRIGHT0, DISENG_SIG_SIGCRCREDREF0, DISENG_SIG_SIGCRCGREENREF0,  
      DISENG_SIG_SIGCRCBLUEREFF0},  
    { DISENG_SIG_EVALCONTROL1, DISENG_SIG_EVALUPPERLEFT1, DISENG_SIG_EVALLOWERRIGHT1, DISENG_SIG_SIGCRCREDREF1, DISENG_SIG_SIGCRCGREENREF1,  
      DISENG_SIG_SIGCRCBLUEREFF1},  
    { DISENG_SIG_EVALCONTROL2, DISENG_SIG_EVALUPPERLEFT2, DISENG_SIG_EVALLOWERRIGHT2, DISENG_SIG_SIGCRCREDREF2, DISENG_SIG_SIGCRCGREENREF2,  
      DISENG_SIG_SIGCRCBLUEREFF2},  
    { DISENG_SIG_EVALCONTROL3, DISENG_SIG_EVALUPPERLEFT3, DISENG_SIG_EVALLOWERRIGHT3, DISENG_SIG_SIGCRCREDREF3, DISENG_SIG_SIGCRCGREENREF3,  
      DISENG_SIG_SIGCRCBLUEREFF3},  
    { DISENG_SIG_EVALCONTROL4, DISENG_SIG_EVALUPPERLEFT4, DISENG_SIG_EVALLOWERRIGHT4, DISENG_SIG_SIGCRCREDREF4, DISENG_SIG_SIGCRCGREENREF4,  
      DISENG_SIG_SIGCRCBLUEREFF4},  
    { DISENG_SIG_EVALCONTROL5, DISENG_SIG_EVALUPPERLEFT5, DISENG_SIG_EVALLOWERRIGHT5, DISENG_SIG_SIGCRCREDREF5, DISENG_SIG_SIGCRCGREENREF5,  
      DISENG_SIG_SIGCRCBLUEREFF5},  
    { DISENG_SIG_EVALCONTROL6, DISENG_SIG_EVALUPPERLEFT6, DISENG_SIG_EVALLOWERRIGHT6, DISENG_SIG_SIGCRCREDREF6, DISENG_SIG_SIGCRCGREENREF6,  
      DISENG_SIG_SIGCRCBLUEREFF6},  
    { DISENG_SIG_EVALCONTROL7, DISENG_SIG_EVALUPPERLEFT7, DISENG_SIG_EVALLOWERRIGHT7, DISENG_SIG_SIGCRCREDREF7, DISENG_SIG_SIGCRCGREENREF7,  
      DISENG_SIG_SIGCRCBLUEREFF7},  
}
```